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Stowable belt buckle

Abstract

A device for engaging a seat belt in a vehicle includes a buckle configured to engage the seat belt, and a base fixedly disposed relative to a seat in the vehicle, where the buckle is moveable relative to the base between a stowed position and an engagement position. The device also includes a latch mechanism including a latch member, and a receiving component fixedly positioned relative to the base. The latch member is configured to be retained in the receiving component when the buckle is manually pushed into the stowed position, and the latch member is configured to be automatically released by manually pushing the buckle when the buckle is in the stowed position.

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Background/Summary

INTRODUCTION

(1) The subject disclosure relates to vehicles, and more particularly, to restraint devices in vehicles.
(2) Many passenger vehicles, particularly those manufactured after 1968, are equipped with passenger restraint systems. Modern vehicles typically feature front restraints that include both lap and shoulder belts. Rear passenger restraints are typically equipped with lap belts, and often include shoulder belts.

SUMMARY

(3) In one exemplary embodiment, a device for engaging a seat belt in a vehicle includes a buckle configured to engage the seat belt, and a base fixedly disposed relative to a seat in the vehicle, where the buckle is moveable relative to the base between a stowed position and an engagement position. The device also includes a latch mechanism including a latch member, and a receiving component fixedly positioned relative to the base. The latch member is configured to be retained in the receiving component when the buckle is manually pushed into the stowed position, and the latch member is configured to be automatically released by manually pushing the buckle when the buckle is in the stowed position.

(4) In addition to one or more of the features described herein, the seat belt is a rear seat belt in a rear occupant compartment of the vehicle.

(5) In addition to one or more of the features described herein, the latch mechanism is selected from a grab latch mechanism and a push-push mechanism.

(6) In addition to one or more of the features described herein, the device is disposed at a recess in the seat, and at least part of the buckle is configured to be located above the recess when the buckle

is in the engagement position.

(7) In addition to one or more of the features described herein, the buckle is configured to be entirely located within the recess when the buckle is in the stowed position.

(8) In addition to one or more of the features described herein, the device includes a biasing mechanism configured to bias the buckle toward the engagement position.

(9) In addition to one or more of the features described herein, the buckle is attached to the base by a pivot pin, and the buckle is rotatable about the pivot pin between the engagement position and the stowed position.

(10) In addition to one or more of the features described herein, the buckle defines a first angle relative to the base when in the stowed position, and the buckle defines a second angle relative to the base when in the engagement position, the first angle being less than the second angle.

(11) In another exemplary embodiment, a method of engaging and disengaging a seat belt in a vehicle includes manually applying a force to a buckle of a buckle assembly, the buckle connected to a base fixedly disposed relative to a seat in the vehicle, where the buckle is moveable relative to the base between a stowed position and an engagement position, the buckle assembly including a latch mechanism having a latch member and a receiving component fixedly positioned relative to the base. Manually applying the force includes pushing the buckle to cause the buckle to move from the engagement position to the stowed position, where the latch member is automatically retained in the receiving component when the buckle reaches the stowed position, and/or pushing the buckle when the buckle is in the stowed position to cause the latch member to be automatically released from the receiving component.

(12) In addition to one or more of the features described herein, the buckle is caused to be returned to the engagement position when the latch member is released from the receiving component.

(13) In addition to one or more of the features described herein, the buckle assembly includes a biasing mechanism configured to bias the buckle toward the engagement position.

(14) In addition to one or more of the features described herein, the seat belt is a rear seat belt in a rear occupant compartment of the vehicle.

(15) In addition to one or more of the features described herein, the buckle is attached to the base by a pivot pin, and the buckle is rotatable about the pivot pin between the engagement position and the stowed position.

(16) In addition to one or more of the features described herein, the buckle defines a first angle relative to the base when in the stowed position, and the buckle defines a second angle relative to the base when in the engagement position, the first angle being less than the second angle.

(17) In addition to one or more of the features described herein, the buckle assembly is disposed at a recess in the seat, and at least part of the buckle is configured to be located above the recess when the buckle is in the engagement position.

(18) In addition to one or more of the features described herein, the buckle is configured to be entirely located within the recess when the buckle is in the stowed position.

(19) In yet another exemplary embodiment, a vehicle system includes a buckle assembly disposed proximate to a rear seat belt in a rear occupant compartment of a vehicle. The buckle assembly includes a buckle configured to engage the rear seat belt, and a base fixedly disposed relative to the rear seat in the vehicle, where the buckle is moveable relative to the base between a stowed position and an engagement position. The buckle assembly also includes a latch mechanism including a latch member and a receiving component fixedly positioned relative to the base, where the latch member is configured to be retained in the receiving component when the buckle is manually pushed into the stowed position. The latch member is configured to be automatically released by manually pushing the buckle when the buckle is in the stowed position, and the buckle is caused to automatically return to the engagement position when the latch member is released from the receiving component.

(20) In addition to one or more of the features described herein, the buckle assembly is disposed at

a recess in the rear seat, at least part of the buckle is configured to be located above the recess when the buckle is in the engagement position.

(21) In addition to one or more of the features described herein, an entirety of the buckle is located within the recess when the buckle is in the stowed position.

(22) In addition to one or more of the features described herein, the buckle is attached to the base by a pivot pin and includes a biasing member configured to bias the buckle toward the engagement position, the buckle is rotatable about the pivot pin between the engagement position and the stowed position, the buckle defines a first angle relative to the base when in the stowed position, and the buckle defines a second angle relative to the base when in the engagement position, the first angle being less than the second angle.

(23) The above features and advantages, and other features and advantages of the disclosure are readily apparent from the following detailed description when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Other features, advantages and details appear, by way of example only, in the following detailed description, the detailed description referring to the drawings in which:

(2) FIG. 1 is a side view of a vehicle, in accordance with an exemplary embodiment;

(3) FIG. 2 is a perspective view of a buckle assembly, in accordance with an exemplary embodiment;

(4) FIG. 3A depicts a portion of a vehicle seat and a buckle assembly in an engagement position, and FIG. 3B depicts a side view of the buckle assembly in the engagement position, in accordance with an exemplary embodiment;

(5) FIG. 4A depicts a portion of a vehicle seat and a buckle assembly in a stowed position, and FIG. 4B depicts a side view of the buckle assembly in the stowed position, in accordance with an exemplary embodiment; and

(6) FIG. 5A depicts a side view of a buckle assembly in a stowed position, FIG. 5B depicts a side view of the buckle assembly during transition to an engagement position, and FIG. 5C depicts a side view of the buckle assembly in the engagement position, in accordance with an exemplary embodiment.

DETAILED DESCRIPTION

(7) The following description is merely exemplary in nature and is not intended to limit the present disclosure, its application or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

(8) In accordance with an exemplary embodiment, methods, devices and systems are provided for restraining a passenger in a vehicle. An embodiment of a passenger restraint system includes a buckle assembly having a buckle and a bracket configured to allow for manually stowing the buckle assembly in a stowed position, and manually releasing the buckle assembly to cause the buckle assembly to automatically return or move to an engagement position (in which the buckle assembly can be engaged with a seat belt).

(9) The bracket includes a base attached to a location below a surface of the seat, and a manual stow and release mechanism (latch mechanism). The manual stow and release mechanism is configured to allow a passenger or user to push (or otherwise apply a force to) the buckle to the stowed position and automatically retain the buckle in the stowed position. The mechanism also allows the user to apply a force to the buckle when the buckle is in the stowed position, which automatically releases the buckle and permits the buckle to return to the engagement position.

(10) Embodiments described herein present numerous advantages and technical effects. The

embodiments provide a simple and effective mechanism for stowing and positioning a seat belt buckle. Current seat belt buckles are placed above a seat cushion surface for ease of use; however, this can cause the buckle to be susceptible to damage (e.g., by cargo being placed in the seat) or cause occupant discomfort when moving across the rear seat. Embodiments address this limitation by allowing buckles to be stowed by a user (e.g., into a pocket in a seat cushion) when not in use, while still being able to self-present at the ideal position for buckling a seat belt when desired by the user (e.g. by the user simply pushing the buckle).

(11) The embodiments described herein are not limited to use with any specific vehicle, device or system, and may be applicable to various devices and systems. For example, embodiments may be used with automobiles, trucks, construction equipment, farm equipment, and/or any other vehicle or machine that includes seat restraints.

(12) FIG. 1 shows a portion of an embodiment of a motor vehicle **10**, which may be a combustion engine vehicle, an electrically powered vehicle (EV) or a hybrid vehicle. The vehicle **10** includes a vehicle body **12** defining, at least in part, a front occupant compartment **14** and a rear occupant compartment **16**. The front occupant compartment includes one or more seats **18**, and the rear occupant compartment includes one or more seats **20**.

(13) The vehicle **10** includes a passenger restraint system **22** configured to be worn by a passenger **24**. The passenger restraint system **22** includes a lap belt **26** and may also include a shoulder belt **28**. The passenger restraint system **22** also includes a buckle assembly **30**.

(14) FIG. 2 depicts an embodiment of the buckle assembly **30**. The buckle assembly **30** is discussed in conjunction with the vehicle **10** and the rear seat **20** for illustration purposes. However, the buckle assembly **30** is not limited to use with the vehicle **10** or similar vehicles.

(15) The buckle assembly **30** includes at least one buckle **32** attached to a bracket **34**. The bracket **34** includes a base **36** designed to be attached or fixedly disposed proximate to a seat, such as the rear seat **20**. The buckle **32** is connected to the base **36** by a rigid connection member **38**.

(16) The buckle assembly **30** is shown in FIG. 2 as including multiple buckles **32**, each of which is connected to the base via a respective connection member **38**. This configuration may be used so as to accommodate adjacent restraints. Alternatively, the buckle assembly **30** may include a single buckle **32**. Also, as shown in FIG. 2, only one latch mechanism **40** (discussed further herein) may be needed, although more than one latch mechanism **40** may be included (e.g., if the buckles are intended to be independently moveable).

(17) Although the buckle assembly **30** is shown as being in the rear occupant compartment **16**, embodiments are not so limited. For example, the buckle assembly **30** may be part of a front seat **18** and/or other seats or seating arrangements, such as multiple rear seat rows, foldable seats and others.

(18) The buckle assembly **30** is moveable between an engagement position and a stowed position. An “engagement position” is any position or orientation of the buckle **32** that allows the buckle **32** to engage a seat belt. For example, the engagement position is an upright position, which may be a vertical position or an angled position.

(19) A “stowed position” is any position or orientation of the buckle **32** where the buckle **32** is positioned so as not to interfere with use of a seat. For example, a stowed position is a position where the buckle **32** is below a surface of the seat **20** or in a recess.

(20) The bracket **34** also includes the latch mechanism **40**, which includes a latch member or actuator member **42**. The latch member **42** is configured to engage a receptacle or receiving component **44**. For example, as shown in FIG. 2, the latch member **42** is a strike member having a rounded end (or other shape), and the receiving component **44** is a grab latch. The latch member **42** may be attached to the connection member **38** and the receiving component **44** may be attached to the base **36** as shown, or vice versa (i.e., the latch member **42** may be attached to the base **36**, with the receiving component **44** attached to the connection member **38**).

(21) In use, when the buckle **32** is in the engagement position, and the buckle **32** is pushed (e.g.,

rotated downwardly), the latch member **42** engages a grab latch **44** (or is otherwise positioned in the receiving component **44**), and the latch member **42** is retained within the receiving component **44** (e.g., by the grab latch **44**) and the buckle **32** is held in the stowed position. When the buckle **32** is again pushed, the latch member **42** is released from the receiving component **44**.

(22) When the buckle **32** is released from the receiving component **44**, the buckle **32** may be freely moveable (e.g., rotated) to allow the user to manually adjust the buckle's position. In an embodiment, the buckle **32**, when engaged with a seat belt, will automatically adjust to a position that aligns with the seat belt.

(23) The latch mechanism is not limited to any particular mechanism. The latch mechanism can be, for example, a cam latch, a push-to-close or slam latch, a push-push slide mechanism or other suitable mechanism or device.

(24) In an embodiment, the buckle **32** is rotatable relative to the base **36**, such that moving the buckle **32** causes the buckle to rotate about a fixed pivot axis **45**. For example, as shown in FIG. 2, the connection member(s) **38** is/are connected to a pivot pin **46** retained at the base **36**.

(25) In an embodiment, the buckle assembly **30** includes a biasing mechanism, such as a spring **48**, which is configured to bias the buckle **32** toward the engagement position. The biasing mechanism causes the buckle **32** to automatically return to a desired engagement position upon release of the latch member **42** from the receiving component **44**.

(26) The biasing mechanism may include any suitable type of spring (e.g., coil spring, leaf spring, etc.) or biasing member. For example, as shown in FIG. 2, the buckle assembly **30** includes a spring **48** coiled around the pivot pin **46**, which causes the buckle **32** to be automatically returned to the engagement position when a user pushes the buckle **32** causing the buckle **32** to disengage from the receiving component **44**. A mechanical stop **50** may be included to restrict the rotational motion of the buckle **32** and establish the engagement position.

(27) FIGS. 3A-3B and 4A-4B show an example of the buckle assembly **30** as positioned proximate to the rear seat **20**. The rear seat **20** includes a recess **52**. The buckle assembly **30** is disposed in the recess **52**, where the base **36** is attached to the structure of the vehicle **10** and fixed with respect to the seat **20**.

(28) The buckle **32** in this example is movable between the engagement position and the stowed position by rotation about the pivot pin **46**. The positions of the buckle **32** are defined in this example, by an angle with respect to a vertical direction V and a horizontal direction H. Movement of the buckle **32** is not limited to rotation, as the buckle can be moved in other ways, such as linearly (e.g., by sliding or otherwise moving along a straight direction).

(29) FIG. 3A shows a portion of the rear compartment **16** of the vehicle **10**, and the buckle assembly **30** in the engagement position. FIG. 3B is a side view of the buckle assembly **30** in the engagement position.

(30) As shown in FIG. 3A, when the buckle assembly **30** is in the engagement position, the buckle **32** is positioned so that at least a portion of the buckle **32** is above a seat surface **19** and can be easily connected to a seat belt. In this example, as shown in FIG. 3B, the buckle assembly **30** defines a first angle $\theta 1$ (e.g., 70-80 degrees), however it is to be understood that the first angle may be any suitable angle (including vertical or 90 degrees).

(31) FIG. 4A shows a portion of the rear compartment **16** of the vehicle **10**, and the buckle assembly **30** in the stowed position. FIG. 4B is a side view of the buckle assembly **30** in the stowed position.

(32) As shown in FIG. 4A, when the buckle assembly **30** is in the stowed position, the buckle assembly **30** is held at an angle such that the entire buckle **32** is below the surface **19** of the seat **20**. For example, the buckle assembly **30** defines a second angle $\theta 2$ (e.g., 20-30 degrees), as shown in FIG. 4B. The angle $\theta 2$ may be any suitable angle (including horizontal or zero degrees). As also shown in FIG. 4B, a user **54** may manually push the buckle **32** to cause the buckle **32** to rotate (as represented by arrow R) into the receiving component **44**, and thereby retain or "click" the buckle

32 in place in the stowed position. When the user **54** wishes to use the seat belt **26** and/or shoulder belt **28**, another “click” or push on the buckle **32** releases the buckle **32** and the buckle assembly automatically returns to the engagement position.

(33) FIGS. **5A**, **5B**, **5C** and **6** depict an embodiment of the buckle assembly **30**, in which the bracket **34** includes a push-push mechanism **60**. The buckle assembly **30** is discussed in conjunction with the vehicle **10** and the rear seat **20** for illustration purposes. However, the buckle assembly **30** is not limited to use with the vehicle **10** or similar vehicles.

(34) The push-push mechanism **60** includes an articulated latch member **62** that is attached to the connection member **38**, and includes a guide component **64** at an end of the latch member **62**. The guide component **64** is disposed in a guide track **66** of a receiving component **68** that is attached to or otherwise fixedly positioned relative to the base **36**.

(35) The guide track **66** forms a loop along which the guide component **64** can travel. The guide track **66** includes a rounded end **70** and a notch portion **72**. The notch portion **72** functions to retain the guide component **64** at a fixed location and thereby keep the connection member **38** at a fixed position. Pushing the connection member **38** or the buckle **32** causes the guide component **64** to move away from the notch portion **72** and allows the connection member **38** to transition between the stowed position and the engagement position.

(36) FIG. **5A** shows the buckle assembly **30** in the stowed position, in which the guide component **64** is retained in the notch portion **72**. A biasing member such as a spring (not shown), biases the connection member **38** toward the engagement position.

(37) To transition to the engagement position, a user applies a force by pushing down so that the connection member **38** and the buckle **32** rotate in the direction shown by arrow **D**, as shown FIG. **5B**. Pushing causes the guide component **64** to move toward an apex **74** of the guide track **66**.

(38) The user subsequently releases the force, and the biasing member causes the connection member **38** to move toward the engagement position, and the guide component **64** moves along path **P** toward the rounded end **70**. When the connection member **38** reaches the engagement position, the guide component **64** is located at the rounded end **70**, as shown in FIG. **5C**. The connection member **38** and the buckle **32** are held in the engagement position by the biasing member. The user can push the buckle **32** again to put the buckle in the stowed position.

(39) Embodiments described herein can be used with preexisting seatbelt buckles with minimal re-design of the buckle brackets. The embodiments can be also used as an optional or luxury feature for high end vehicles to improve the appearance of rear row seats.

(40) The terms “a” and “an” do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item. The term “or” means “and/or” unless clearly indicated otherwise by context. Reference throughout the specification to “an aspect”, means that a particular element (e.g., feature, structure, step, or characteristic) described in connection with the aspect is included in at least one aspect described herein, and may or may not be present in other aspects. In addition, it is to be understood that the described elements may be combined in any suitable manner in the various aspects.

(41) When an element such as a layer, film, region, or substrate is referred to as being “on” another element, it can be directly on the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” another element, there are no intervening elements present.

(42) Unless specified to the contrary herein, all test standards are the most recent standard in effect as of the filing date of this application, or, if priority is claimed, the filing date of the earliest priority application in which the test standard appears.

(43) Unless defined otherwise, technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this disclosure belongs.

(44) While the above disclosure has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents

may be substituted for elements thereof without departing from its scope. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiments disclosed, but will include all embodiments falling within the scope thereof.

Claims

1. A device for engaging a seat belt in a vehicle, comprising: a buckle configured to engage the seat belt; a base fixedly disposed relative to a seat in the vehicle, wherein the buckle is moveable relative to the base between a stowed position and an engagement position; a biasing mechanism configured to apply a biasing force to the buckle, the biasing force urging the buckle toward the engagement position; and a latch mechanism including a latch member, and a receiving component fixedly positioned relative to the base, wherein the latch member is configured to be retained in the receiving component when the buckle is manually pushed in a first direction into the stowed position, and the latch member is configured to be automatically released from the receiving component by manually pushing the buckle in the first direction when the buckle is in the stowed position, the biasing force causing the buckle to automatically return to the engagement position upon release of the buckle, wherein the latch mechanism includes a grab latch mechanism, the grab latch mechanism including a strike member and a grab latch, the grab latch configured to retain the strike member when the buckle is in the stowed position, and release the strike member and the buckle from the stowed position when the buckle is pushed in the first direction.
2. The device of claim 1, wherein the seat belt is a rear seat belt in a rear occupant compartment of the vehicle.
3. The device of claim 1, wherein the latch mechanism includes a mechanical stop configured to restrict a motion of the buckle and establish the engagement position.
4. The device of claim 1, wherein the device is disposed at a recess in the seat, and at least part of the buckle is configured to be located above the recess when the buckle is in the engagement position.
5. The device of claim 4, wherein the buckle is configured to be entirely located within the recess when the buckle is in the stowed position.
6. The device of claim 1, wherein the latch mechanism is a push-push mechanism, the push-push mechanism includes an articulated member, the receiving component includes a guide track configured to receive an end of the articulated member, and manually pushing the buckle causes the end to move along the guide track and allow the buckle to automatically return to the engagement position.
7. The device of claim 1, wherein the buckle is attached to the base by a pivot pin, and the buckle is rotatable about the pivot pin between the engagement position and the stowed position.
8. The device of claim 7, wherein the buckle defines a first angle relative to the base when in the stowed position, and the buckle defines a second angle relative to the base when in the engagement position, the first angle being less than the second angle.
9. A method of engaging and disengaging a seat belt in a vehicle, comprising: manually applying a force to a buckle of a buckle assembly, the buckle connected to a base fixedly disposed relative to a seat in the vehicle, wherein the buckle is moveable relative to the base between a stowed position and an engagement position, the buckle assembly including a latch mechanism having a latch member and a receiving component fixedly positioned relative to the base, the buckle assembly including a biasing mechanism configured to apply a biasing force to the buckle, the biasing force urging the buckle toward the engagement position, wherein the latch mechanism includes a grab latch mechanism, the grab latch mechanism including a strike member and a grab latch, the grab latch configured to retain the strike member when the buckle is in the stowed position, and release

the strike member and the buckle from the stowed position when the buckle is pushed in the first direction, and wherein manually applying the force includes at least one of: pushing the buckle in a first direction to cause the buckle to move from the engagement position to the stowed position, wherein the latch member is automatically retained in the receiving component when the buckle reaches the stowed position; and pushing the buckle in the first direction when the buckle is in the stowed position to cause the latch member to be automatically released from the receiving component, the biasing force causing the buckle to automatically return to the engagement position upon release of the buckle.

10. The method of claim 9, wherein the latch mechanism includes a mechanical stop configured to restrict a motion of the buckle and establish the engagement position.

11. The method of claim 10, wherein the latch mechanism is a push-push mechanism, the push-push mechanism includes an articulated member, the receiving component includes a guide track configured to receive an end of the articulated member, and manually pushing the buckle causes the end to move along the guide track and allow the buckle to automatically return to the engagement position.

12. The method of claim 9, wherein the seat belt is a rear seat belt in a rear occupant compartment of the vehicle.

13. The method of claim 9, wherein the buckle is attached to the base by a pivot pin, and the buckle is rotatable about the pivot pin between the engagement position and the stowed position.

14. The method of claim 13, wherein the buckle defines a first angle relative to the base when in the stowed position, and the buckle defines a second angle relative to the base when in the engagement position, the first angle being less than the second angle.

15. The method of claim 9, wherein the buckle assembly is disposed at a recess in the seat, and at least part of the buckle is configured to be located above the recess when the buckle is in the engagement position.

16. The method of claim 15, wherein the buckle is configured to be entirely located within the recess when the buckle is in the stowed position.

17. A vehicle system comprising: a buckle assembly disposed proximate to a rear seat belt in a rear occupant compartment of a vehicle, the buckle assembly including: a buckle configured to engage the rear seat belt; a base fixedly disposed relative to the rear seat in the vehicle, wherein the buckle is moveable relative to the base between a stowed position and an engagement position; a biasing mechanism configured to apply a biasing force to the buckle, the biasing force urging the buckle toward the engagement position; and a latch mechanism including a latch member and a receiving component fixedly positioned relative to the base, wherein the latch member is configured to be retained in the receiving component when the buckle is manually pushed in a first direction into the stowed position, the latch member is configured to be automatically released from the receiving component by manually pushing the buckle in the first direction when the buckle is in the stowed position, and the buckle is caused by the biasing force to automatically return to the engagement position when the latch member is released from the receiving component, wherein the latch mechanism includes a grab latch mechanism, the grab latch mechanism including a strike member and a grab latch, the grab latch configured to retain the strike member when the buckle is in the stowed position, and release the strike member and the buckle from the stowed position when the buckle is pushed in the first direction.

18. The vehicle system of claim 17, wherein the buckle assembly is disposed at a recess in the rear seat, at least part of the buckle is configured to be located above the recess when the buckle is in the engagement position.

19. The vehicle system of claim 18, wherein an entirety of the buckle is located within the recess when the buckle is in the stowed position.

20. The vehicle system of claim 17, wherein the buckle is attached to the base by a pivot pin and includes a biasing member configured to bias the buckle toward the engagement position, the

buckle is rotatable about the pivot pin between the engagement position and the stowed position, the buckle defines a first angle relative to the base when in the stowed position, and the buckle defines a second angle relative to the base when in the engagement position, the first angle being less than the second angle.
